

PRODUCT BACKLOG: group 5.

→ **Product Goal:** Build a multi-agent rescue simulation where agents explore a damaged area, detect targets, communicate useful information, learn from the environment, and improve rescue strategy over time.

→ **Product Backlog:**

Priority	Item	Reason / Value	Estimate
1	Set up development infrastructure	The team needs GitLab, coding language, development environment, and base project structure before implementation can start.	18h
2	Integrate CI and automated testing	Tests should run automatically when new code is uploaded, so the project remains stable during development.	6h
3	Create damaged-area grid environment	The simulator needs a grid world with blocked cells, walls, valid positions, obstacles, and reproducible random generation.	9h
4	Add Target A and Target B spawn mechanics	Rescue scenarios need different target types instead of generic targets.	6h
5	Implement agent movement model	Agents must move through the environment while avoiding walls and obstacles.	16h
6	Add central sensor and basic communication	The agent must be able to ask the central sensor for observations, and the system needs basic information exchange for later coordination.	18h
7	Add simple visual feedback and simulation loop	The team needs to run and demonstrate a basic scenario with text output and metrics.	16h
8	Implement autonomous exploration baseline	Agents should start exploring without manual movement decisions.	18h
9	Define state-space data structure and reward logic	The learning system needs a clear representation of states, actions, transitions, and rewards.	15h
10	Implement single-agent rescue learning logic and evaluate	A single agent should learn or improve behavior before adding multiple agents, the team should evaluate different scenarios.	30h
11	Support multiple agent starting positions	The simulator must be able to place a defined set of agents at their starting positions.	10h

12	Extend agent-sensor communication for multiple agents	Multiple agents need sensor observations of obstacles, targets, and other agents to support cooperative rescue behavior.	12h
13	Add central multi-agent reinforcement learning architecture	A central server should organize and optimize exploration and rescue strategies.	30h
14	Add agent-to-agent communication with radius limit	Agents should only communicate with nearby agents, as required for distributed coordination.	15h
15	Add distributed multi-agent learning architecture	Agents should optimize behavior as a society without relying on a central optimization server.	24h
16	Expand reward logic and strategy evaluation for multi-agent	The system needs rewards and metrics to compare different strategies.	12h
17	Validate results with test scenarios	The team needs reliable evaluation and performance estimates across different scenarios.	8h
18	Optimize precision/performance	Improve the quality of predictions or rescue decisions based on validation results.	12h
19	Integrate uncertainty in movement and sensor readings	Real-world conditions require uncertain movement and sensor readings.	12h
20	Build graphical simulator / visual interface	The official project asks for a graphical simulator showing the damaged area, agents, obstacles, targets, and rescue process.	18h
21	Prepare final live demo and presentation	The product must be understandable and demonstrable for final delivery.	18h

Colours:

Sprint1-purple

Sprint2-yellow

Sprint3-blue

Sprint4-red

Sprint5-green

FinalSprint-pink